Lorena Marcos

Machine Learning & AI | Applied Data Science Student

Madrid, Spain · Available from February 2027, 20–30 h/week (afternoons)
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SUMMARY

Applied Data Science undergraduate at Universidad Complutense de Madrid, focused on machine learning, AI engineering and building end-to-end software products. Hands-on experience with Python, SQL, predictive modeling and statistical analysis, plus a full-stack platform whose backend is deployed to production for a real event pilot. Seeking a paid, part-time ML/AI role from February 2027, with the goal of moving to full-time after graduation.

EDUCATION

BSc in Applied Data Science — Universidad Complutense de Madrid 2023 – Present

- Relevant coursework: Machine Learning, Supervised & Unsupervised Learning, Data Mining, Time Series, Statistical Inference, Linear Algebra, Discrete Mathematics.

PROJECTS

Ride-Sharing Cancellation Prediction — Machine Learning 2026

XGBoost · R · tidyverse · dbscan · Quarto · testthat

- Built a leakage-safe XGBoost classifier to predict customer cancellations on 150,000 ride-sharing bookings: temporal train/test split, validation-only tuning, and logistic-regression and no-skill baselines.
- Showed that booking-time features carry no signal (ROC-AUC 0.50) and traced the apparent lift from vehicle arrival time (PR-AUC 0.48 vs 0.07 baseline) to an artefact of synthetically generated data.
- Handled structural missingness (MNAR) and duplicate IDs; detected anomalies with IQR and Local Outlier Factor; feature preparation covered by automated tests.

github.com/Lorenadata/ncr-ride-sharing-eda

ESCUDO MADRID — Wildfire Prevention Planning Platform (DATAMAD 2026 entry) 2026

Python · FastAPI · PostgreSQL/PostGIS · OR-Tools (CP-SAT) · Next.js · MapLibre · Docker · GitHub Actions

- Open platform that turns Madrid open data (11 regional datasets, official risk cartography, AEMET forecasts, EFFIS burnt areas) into preventive action plans for municipalities, with checksummed, reproducible pipelines.
- Deterministic budget optimizer (integer programming, CP-SAT) compared against two baselines; built by directing Claude Code agents from a written spec, with automated tests, CI and a public demo.

lorenadata-escudo-madrid.hf.space · github.com/Lorenadata/escudo-madrid-incendios

RaceTrack — Sports Event Management Platform (independent project) 2026 – Present

Python · FastAPI · SQLAlchemy · Alembic · PostgreSQL/PostGIS (Supabase) · Redis · React PWA · Stripe · Railway

- Building a full-stack platform for popular races and charity walks: participant registration, payments, maps and push notifications.
- Designed and deployed the REST API backend to production: database migrations, JWT authentication, Stripe webhooks, QR-based bib verification and geospatial data with PostGIS and MapLibre.
- Pilot with a real charity march of 12,000–15,000 participants, working directly with the event organizer.

github.com/Lorenadata/racetrack

Fiscalis — Database for a Tax Management SaaS 2026

SQL · MySQL 8 · Relational design

- Designed the 11-table relational model of an MVP for Spanish freelancers (fiscal profile, clients, invoices, expenses, quarterly tax filings and deadline alerts), with domain rules enforced in the schema.

github.com/Lorenadata/fiscalis-db

Income Inequality & Child Poverty in Spain — Official Microdata 2026

R · dplyr · Hmisc · ggplot2 · Quarto · INE ECV 2024

- Survey-weighted analysis (RB050) of tenants vs. landlords and child poverty (AROPE): weighted percentiles and P90/P10 ratio; published as reproducible reports.

lorenadata.github.io/ECV-rental-market

EXPERIENCE

Solar Energy Hackathon — Predictive Pricing Model 2024

- Built a photovoltaic energy price prediction model in a multidisciplinary team in under 48 hours (Python, feature engineering); presented to a technical panel.

SKILLS

Programming	Python (pandas, NumPy), SQL, R (tidyverse, ggplot2, Quarto)
Machine Learning	scikit-learn, XGBoost, statsmodels; classification, regression (GLM), supervised & unsupervised learning
Statistics	Statistical inference, survey weighting, outlier detection (LOF, IQR), missing-data analysis, time series
AI	Anthropic Claude API for workflow automation; spec-driven development with Claude Code agents
Backend & Data	FastAPI, REST APIs, PostgreSQL, PostGIS, MySQL, MongoDB, SQLAlchemy, Alembic, Redis
Tools	Git, GitHub Actions, Docker, Railway, Jupyter, React, Next.js
Languages	Spanish (native), English (B2)